CS 301: High-Performance Computing

Assignment 05

Parallel insertion and deletion in Mover

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March 31, 2026

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1 Problem Description

This assignment extends the particle interpolation problem by introducing dynamic particle updates: deletion and insertion. In previous labs, particles were confined to the domain. In this assignment, particles move freely, and if they exit the 1×1 unit square, they are removed (deleted). To maintain a constant particle count, new particles are randomly inserted to replace the deleted ones. Two implementation strategies are explored:

- **Deferred Insertion (Approach 1):** Particles are moved in a loop, deletions are tracked, and new particles are inserted at the end of the iteration to fill the voids.
- **Immediate Replacement (Approach 2):** When a particle exits the domain, it is immediately replaced by a new particle at a random position within the domain during the same loop iteration.

2 Hardware Details

2.1 Hardware Details for LAB207 PCs

- **Model name:** 12th Gen Intel(R) Core(TM) i5-12500
- **Architecture:** x86_64
- **CPU(s):** 12 (6 Cores, 12 Threads)
- **Memory:** 16 GB (Approx.)

2.2 Hardware Details for HPC Cluster (Node gics0)

- **Model name:** Intel(R) Xeon(R) CPU E5-2620 v3 @ 2.40GHz
- **Architecture:** x86_64
- **CPU(s):** 24 (2 Sockets, 12 Cores, 24 Threads)
- **Memory:** 128 GB (Approx.)

3 Implementation and Methodology

Approach 1 (Deferred Insertion) marks particles as invalid and performs a repacking/insertion step at the end. Approach 2 (Immediate Replacement) updates positions in-situ. Both are parallelized using OpenMP with thread-safe random number generation.

4 Experiment 01 Results: Scaling with Particle Count

4.1 Approach 1 (Deferred Insertion)

4.1.1 Grid 1 (250x100)

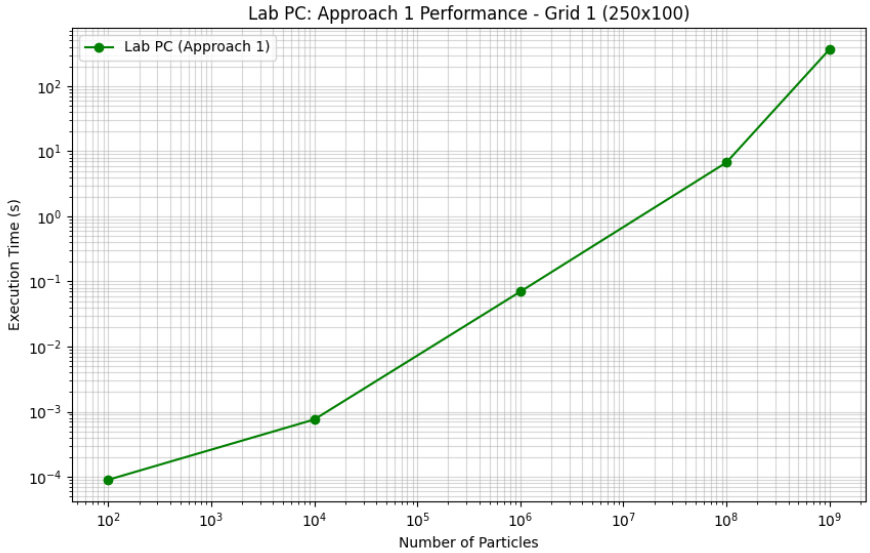

Figure 1: Lab PC Timing (Grid 1)

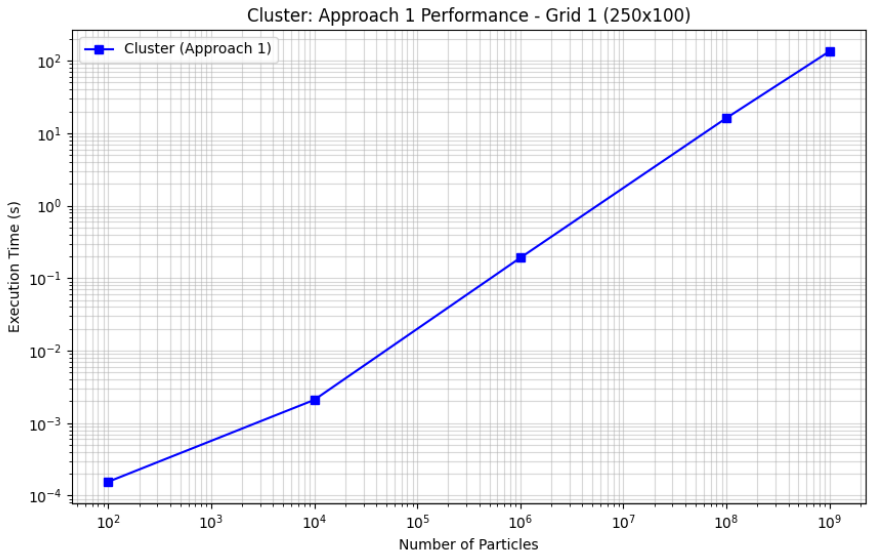

Figure 2: HPC Cluster Timing (Grid 1)

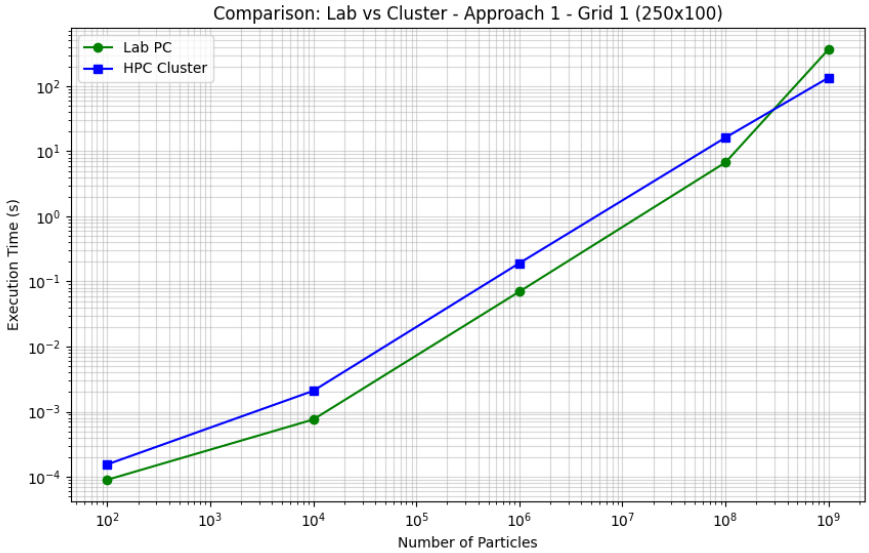

Figure 3: Platform Comparison (Grid 1)

4.1.2 Grid 2 (500x200)

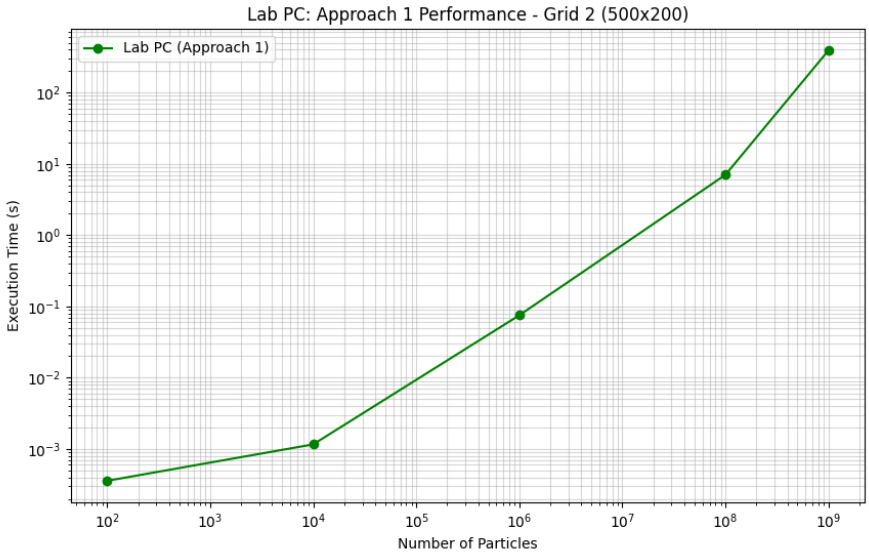

Figure 4: Lab PC Timing (Grid 2)

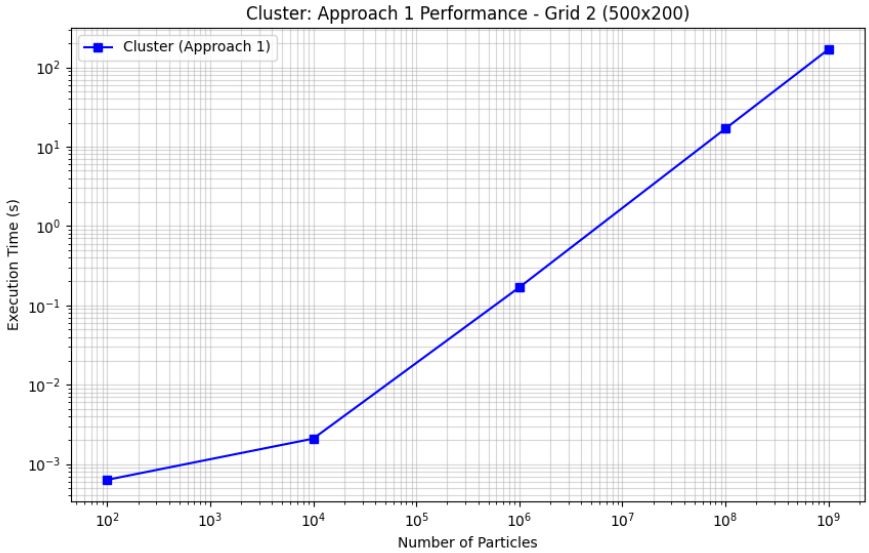

Figure 5: HPC Cluster Timing (Grid 2)

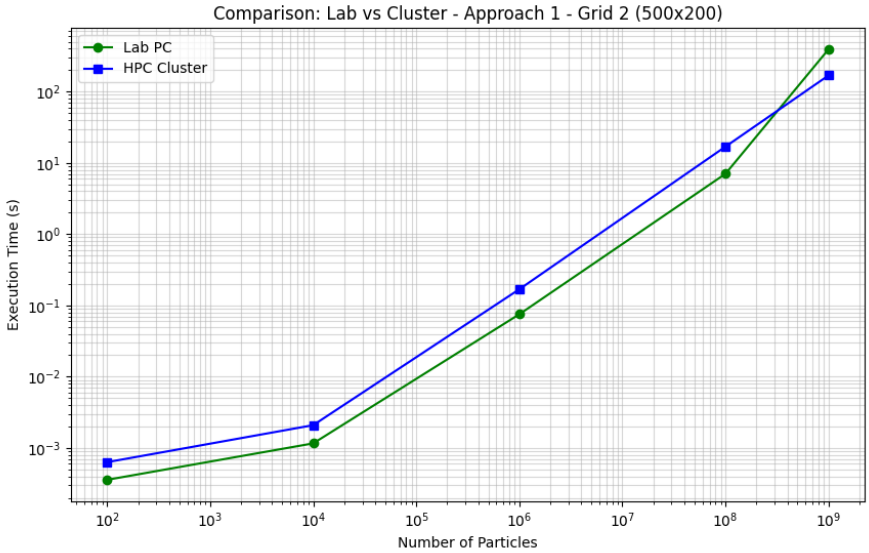

Figure 6: Platform Comparison (Grid 2)

4.1.3 Grid 3 (1000x400)

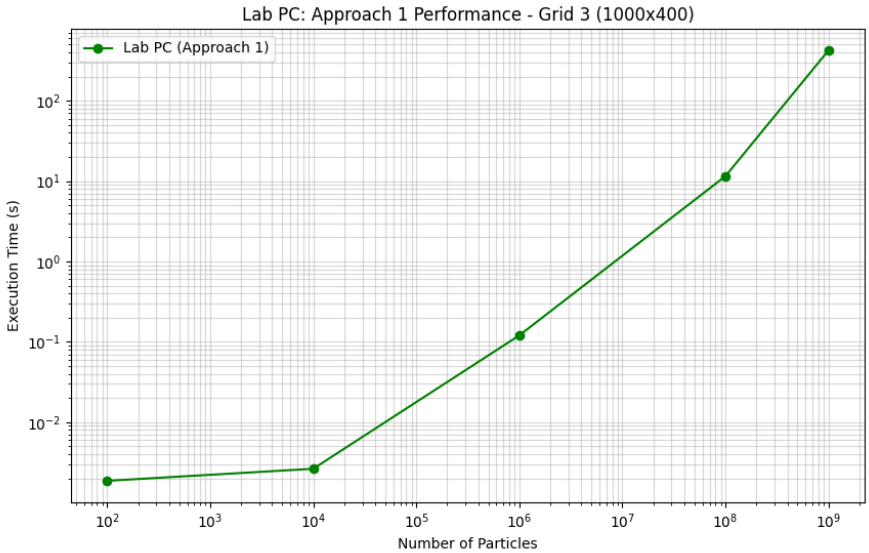

Figure 7: Lab PC Timing (Grid 3)

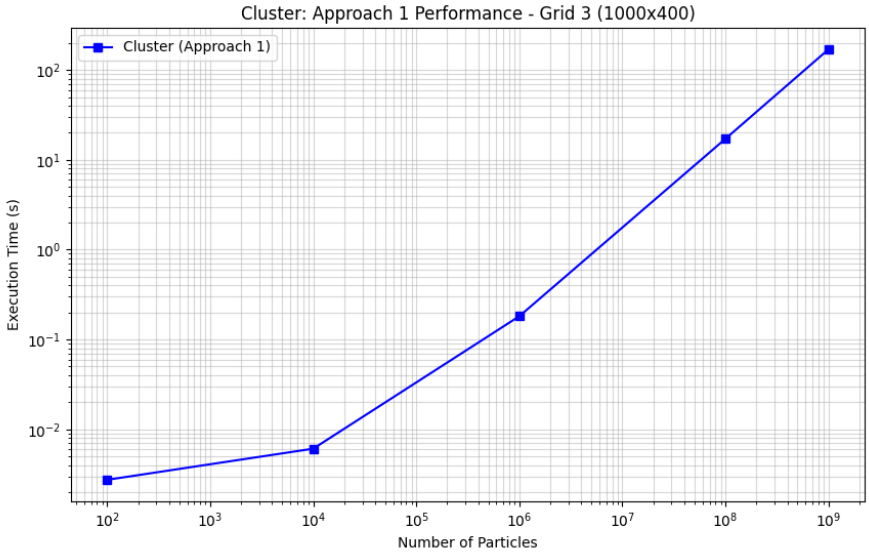

Figure 8: HPC Cluster Timing (Grid 3)

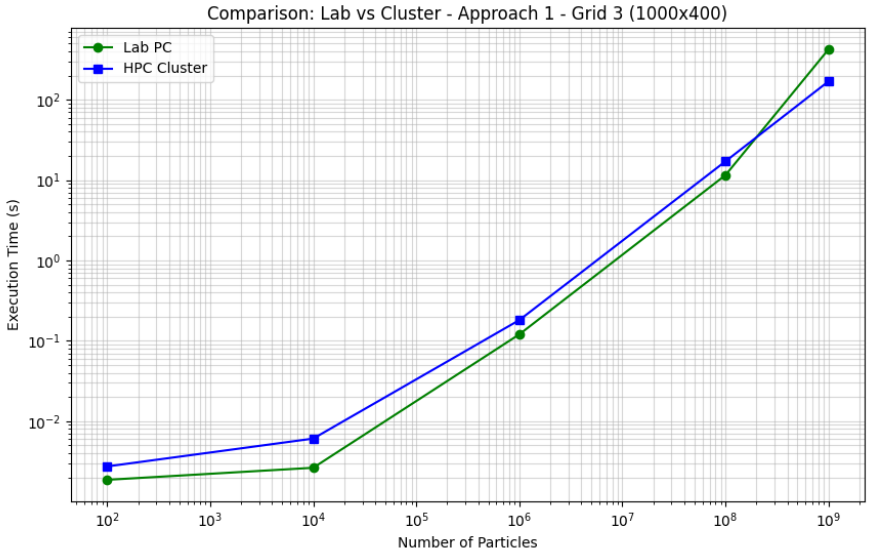

Figure 9: Platform Comparison (Grid 3)

4.2 Approach 2 (Immediate Replacement)

4.2.1 Grid 1 (250x100)

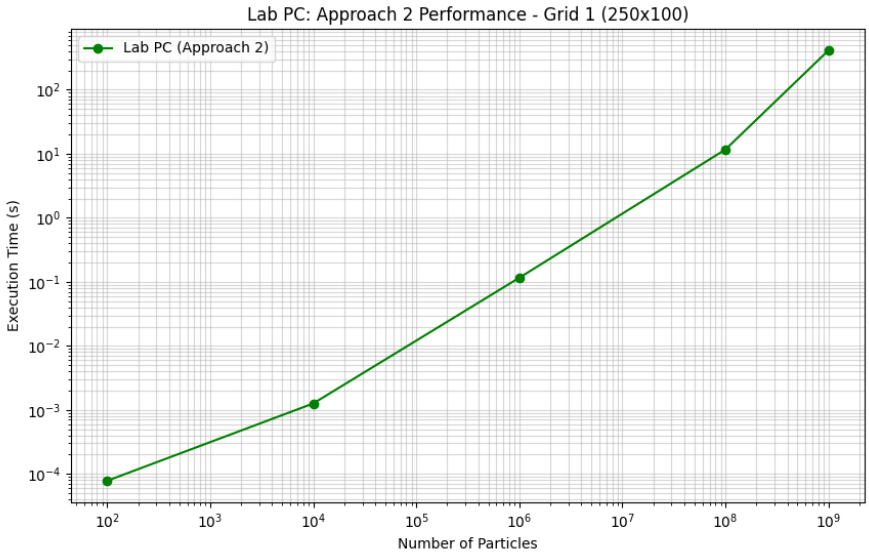

Figure 10: Lab PC Timing (Grid 1)

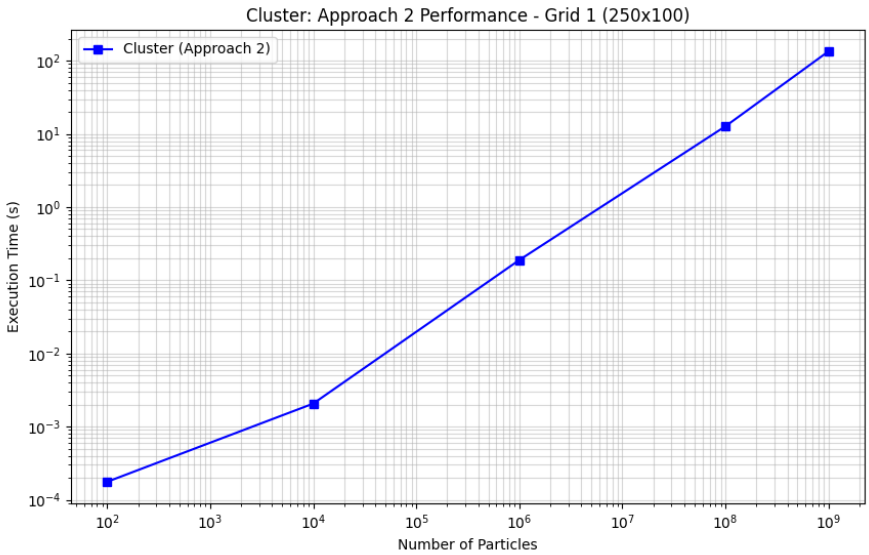

Figure 11: HPC Cluster Timing (Grid 1)

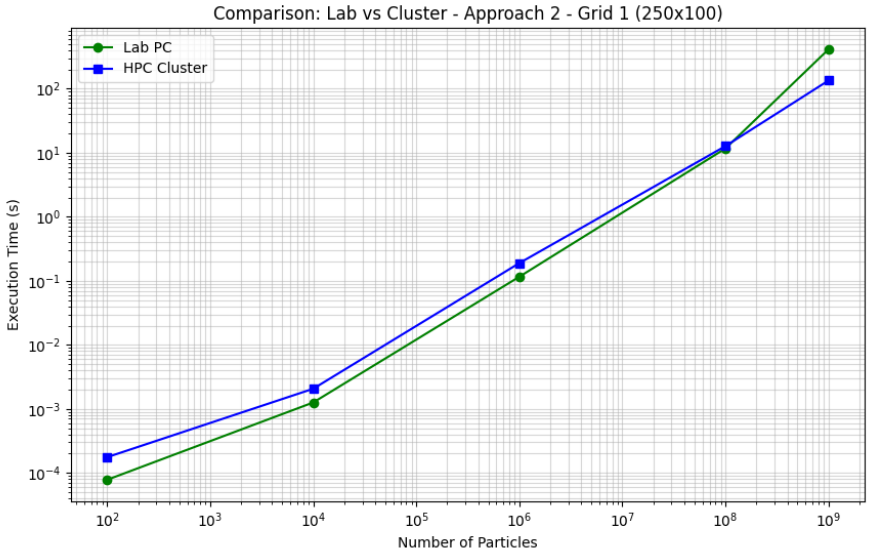

Figure 12: Platform Comparison (Grid 1)

4.2.2 Grid 2 (500x200)

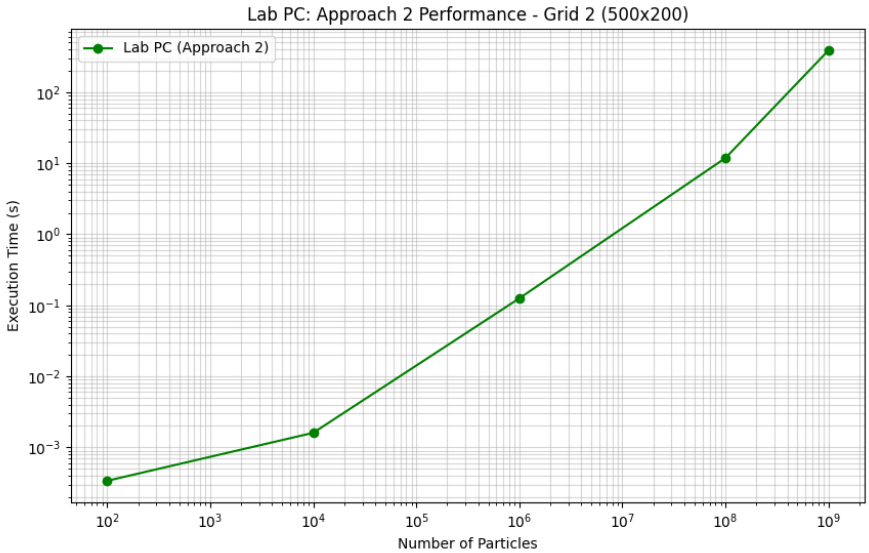

Figure 13: Lab PC Timing (Grid 2)

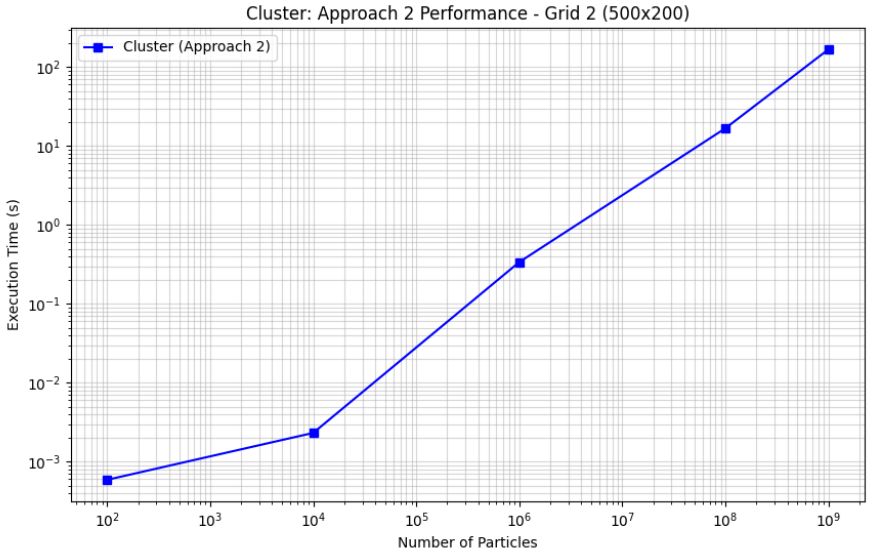

Figure 14: HPC Cluster Timing (Grid 2)

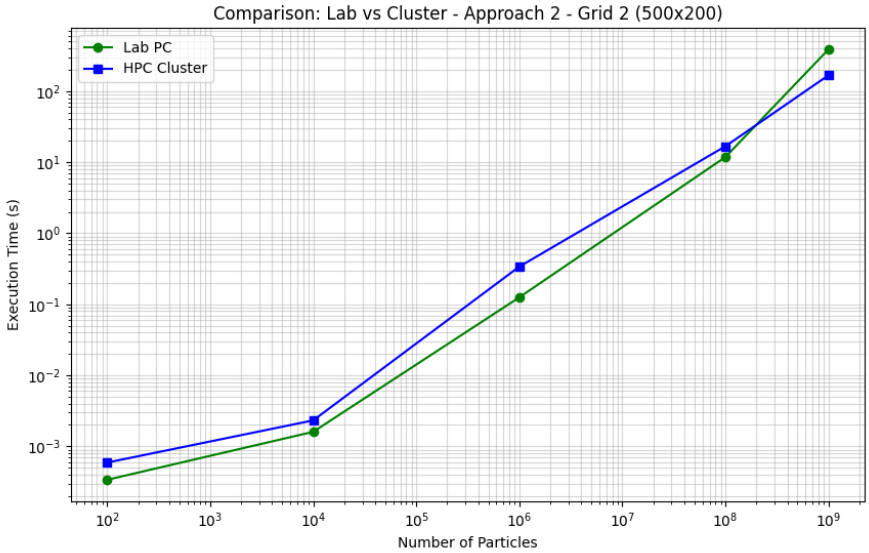

Figure 15: Platform Comparison (Grid 2)

4.2.3 Grid 3 (1000x400)

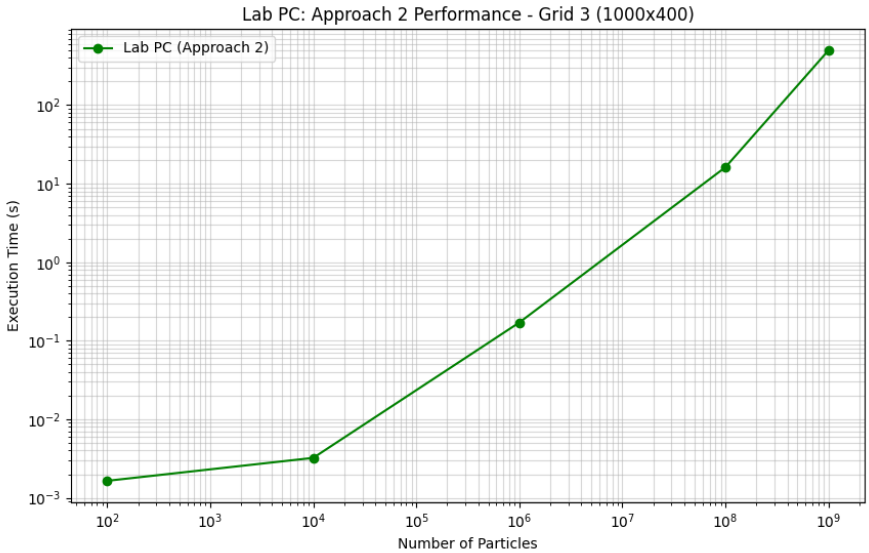

Figure 16: Lab PC Timing (Grid 3)

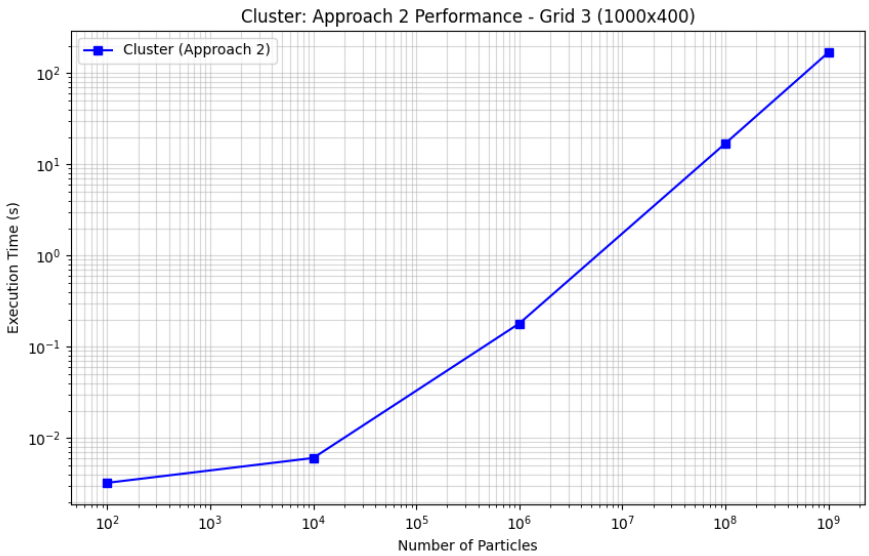

Figure 17: HPC Cluster Timing (Grid 3)

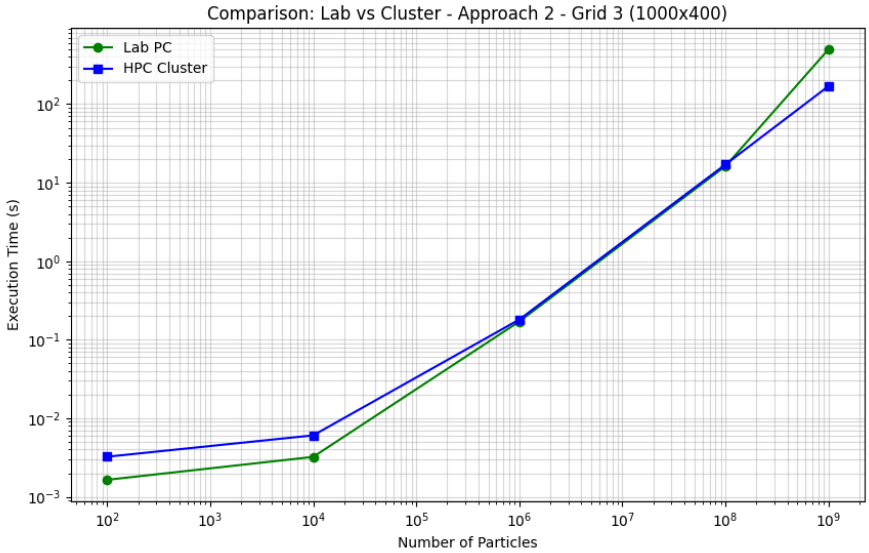

Figure 18: Platform Comparison (Grid 3)

5 Experiment 02 Results: Parallel Scalability

5.1 Approach 1 (Deferred Insertion)

5.1.1 Scalability Results Across Grids

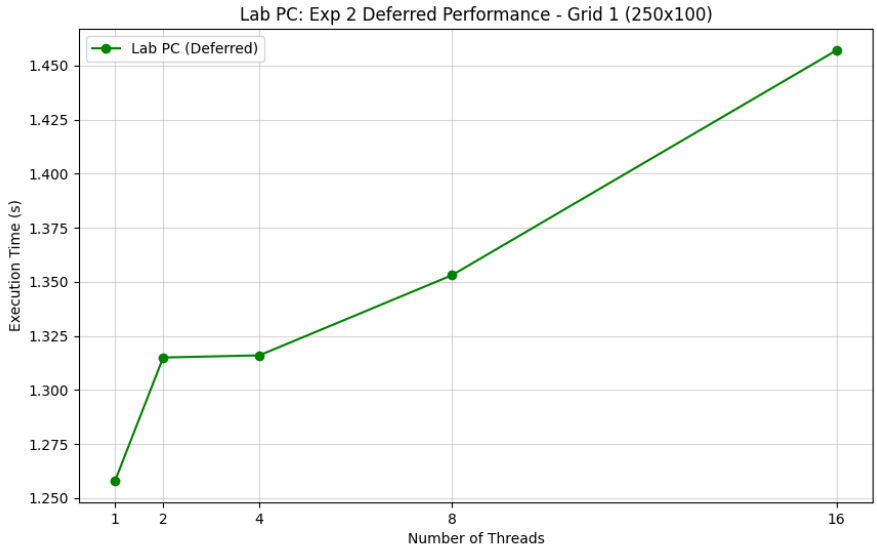

Figure 19: Lab PC Speedup (Grid 1)

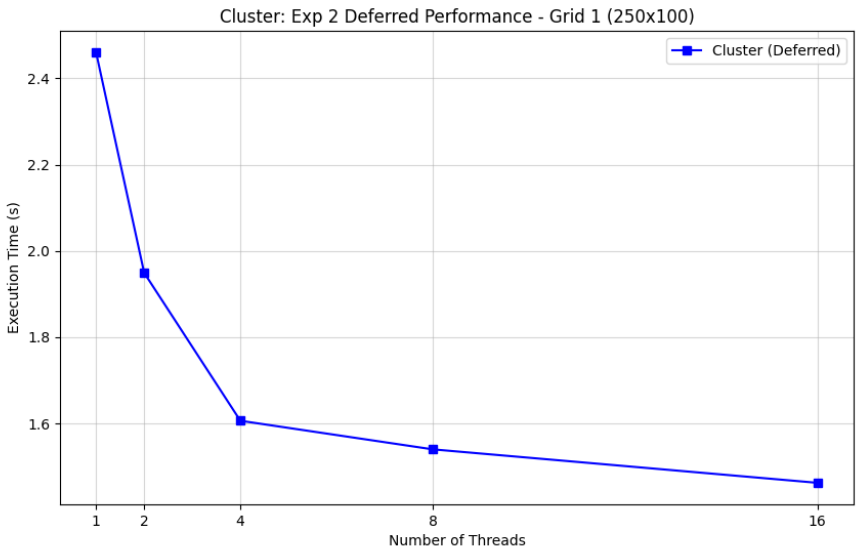

Figure 20: Cluster Speedup (Grid 1)

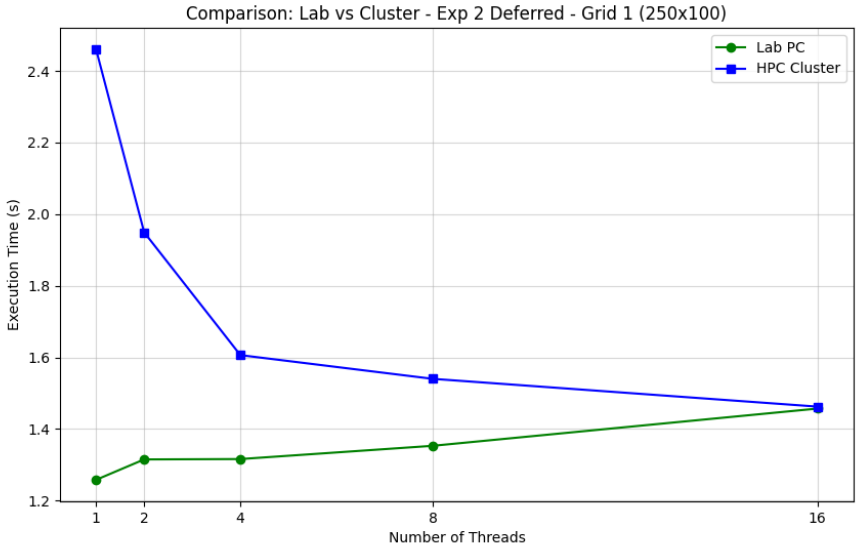

Figure 21: Mover Comparison (Asgn 04 vs 05) - Grid 1

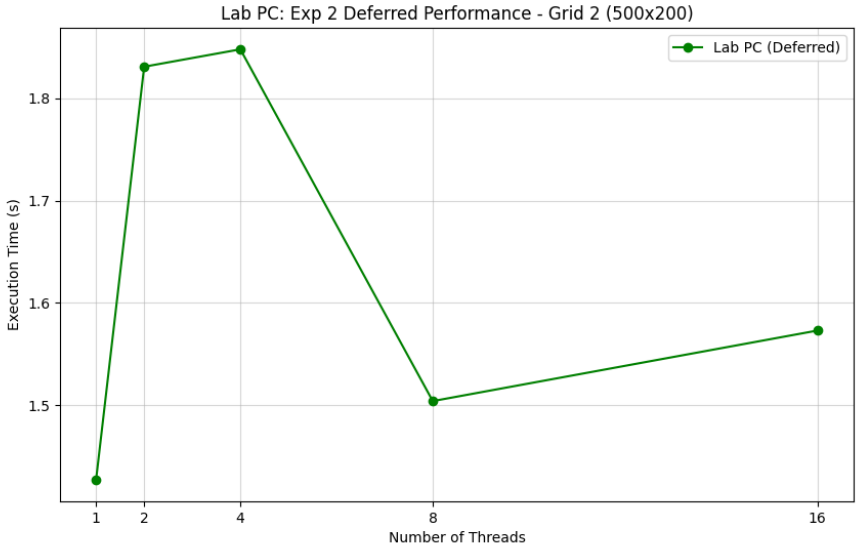

Figure 22: Lab PC Speedup (Grid 2)

Figure 23: Cluster Speedup (Grid 2)

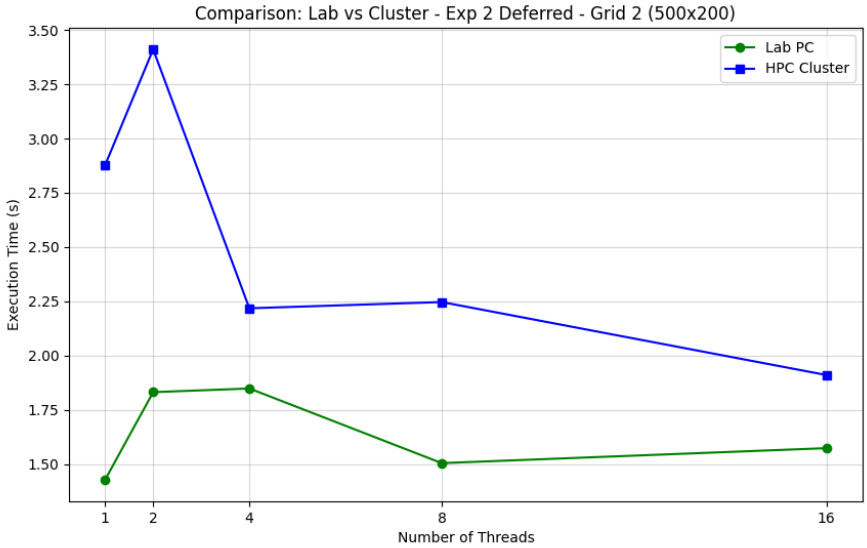

Figure 24: Mover Comparison (Asgn 04 vs 05) - Grid 2

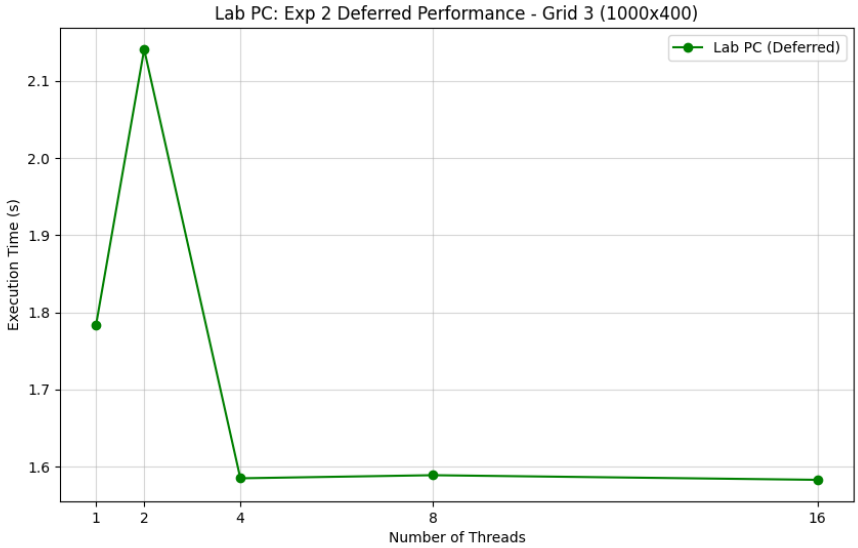

Figure 25: Lab PC Speedup (Grid 3)

Figure 26: Cluster Speedup (Grid 3)

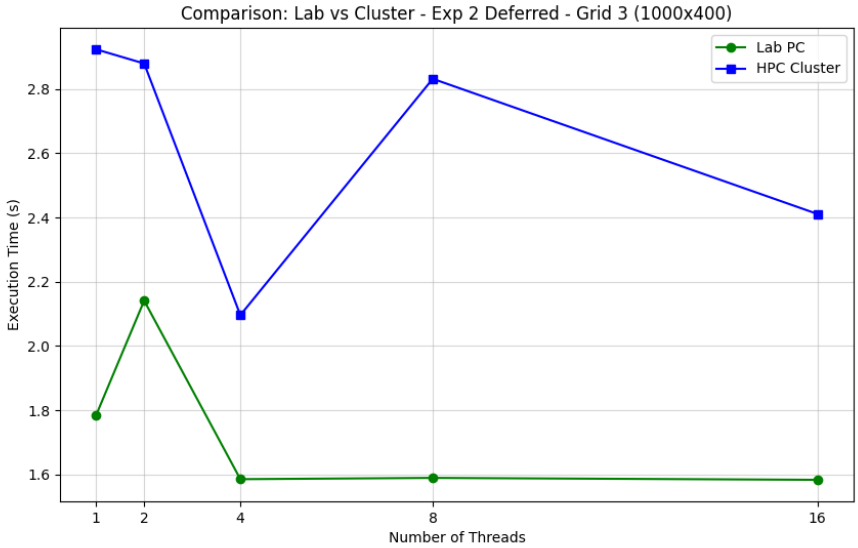

Figure 27: Mover Comparison (Asgn 04 vs 05) - Grid 3

5.2 Approach 2 (Immediate Replacement)

5.2.1 Scalability Results Across Grids

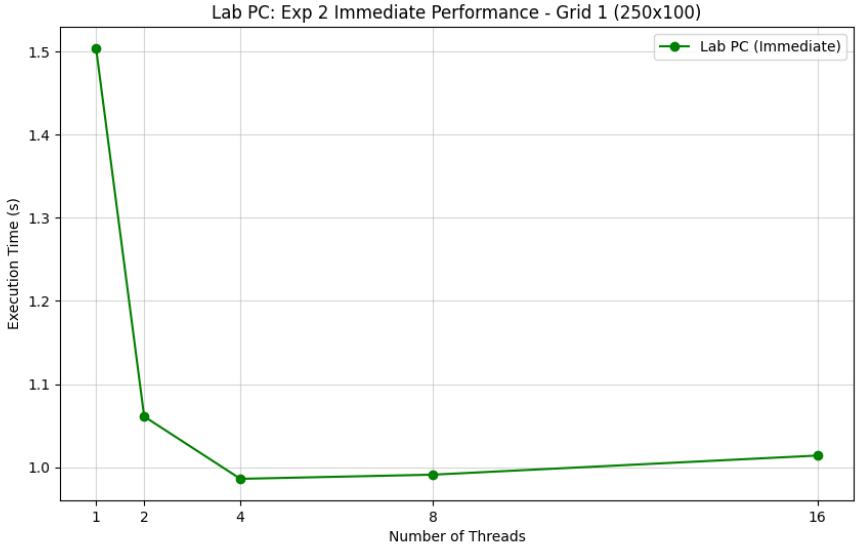

Figure 28: Lab PC Speedup (Grid 1)

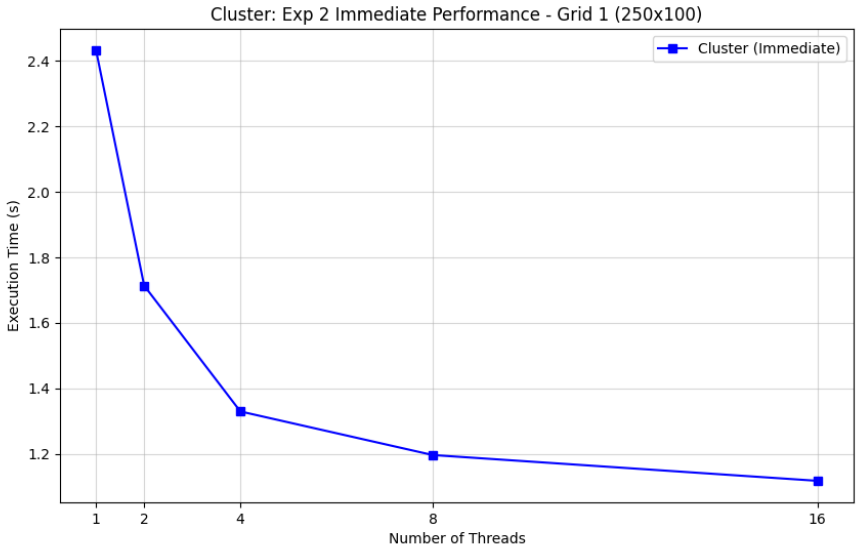

Figure 29: Cluster Speedup (Grid 1)

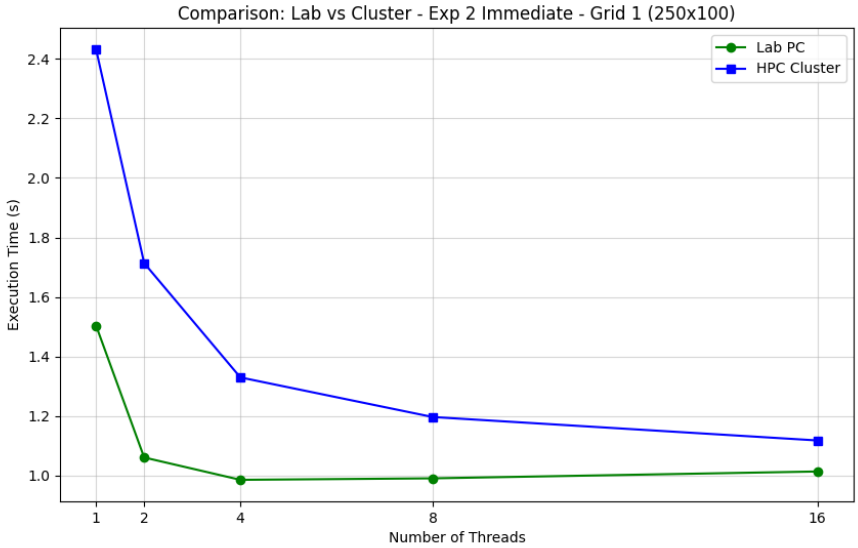

Figure 30: Mover Comparison (Asgn 04 vs 05) - Grid 1

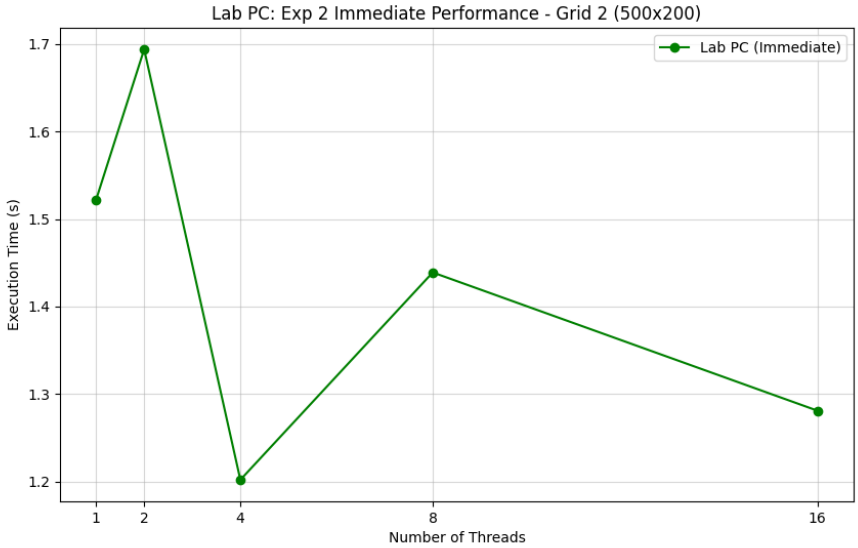

Figure 31: Lab PC Speedup (Grid 2)

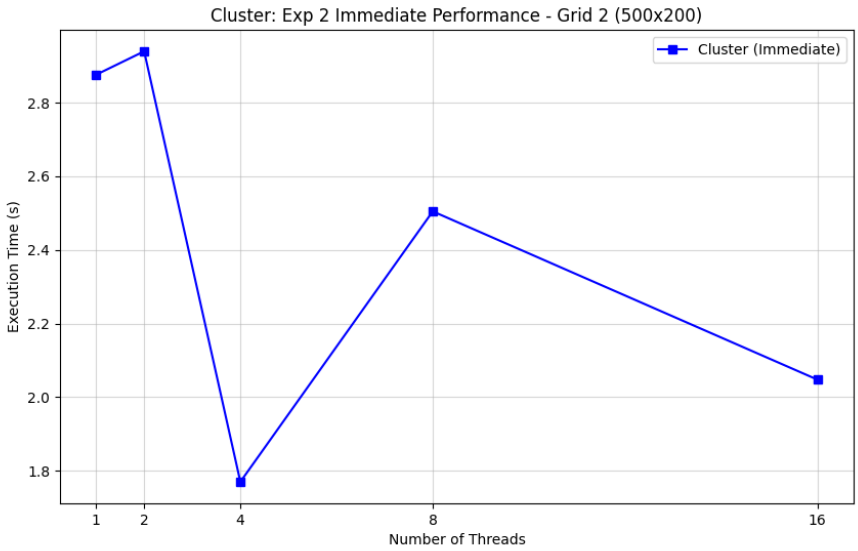

Figure 32: Cluster Speedup (Grid 2)

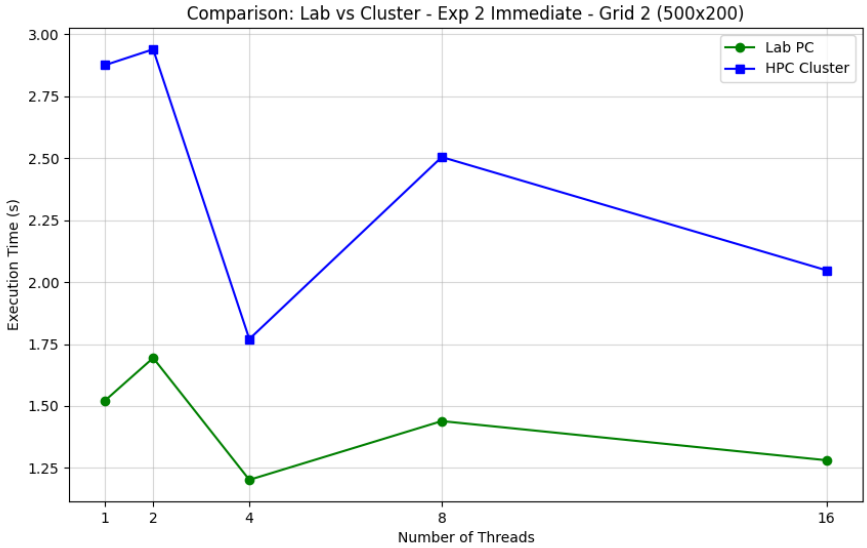

Figure 33: Mover Comparison (Asgn 04 vs 05) - Grid 2

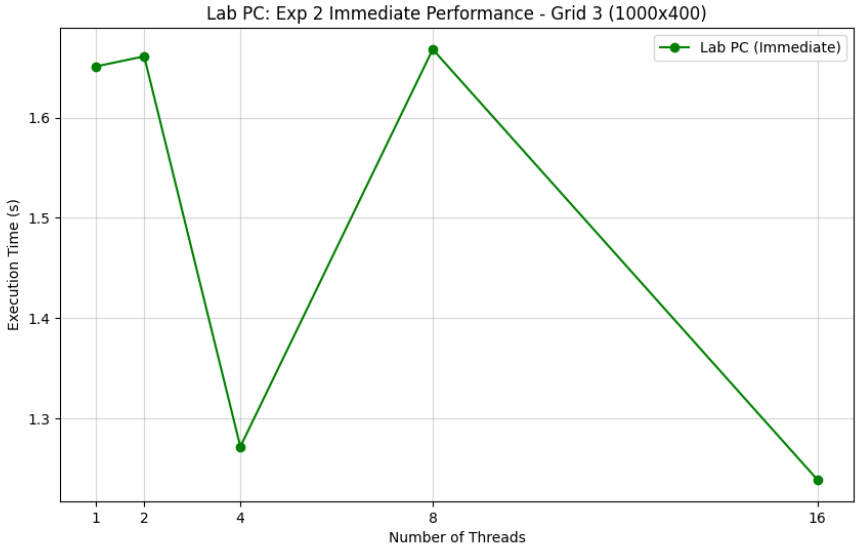

Figure 34: Lab PC Speedup (Grid 3)

Figure 35: Cluster Speedup (Grid 3)

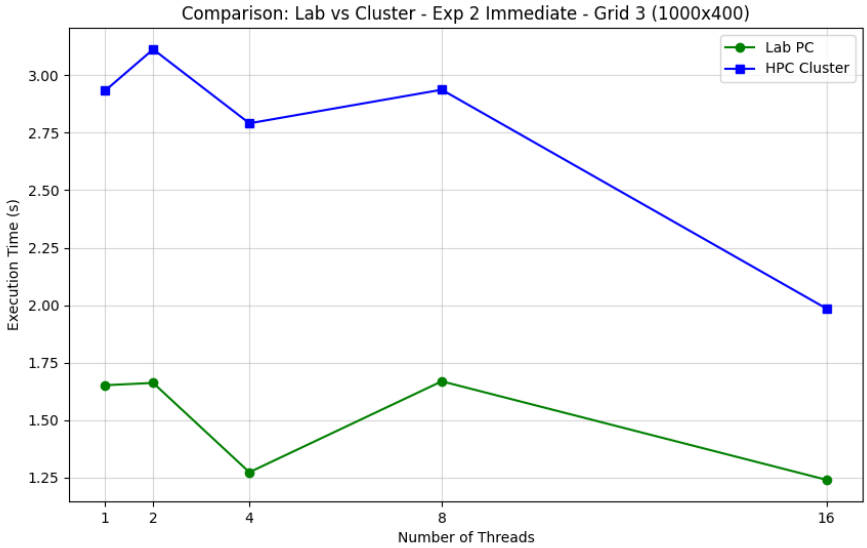

Figure 36: Mover Comparison (Asgn 04 vs 05) - Grid 3

6 Analysis and Discussion

6.1 Theoretical Time Complexity

The complexity is $O(\text{Maxiter} \times N)$. Insertion and deletion add a small constant overhead per particle.

6.2 Memory and Computation Analysis

Particles	Memory (Points)	Memory (Grid)	Total
10^6	16 MB	0.2 MB	16.2 MB
10^8	1.6 GB	0.2 MB	1.6 GB
10^9	16 GB	0.2 MB	16 GB

Table 1: Memory requirements.

6.3 Particles Per Cell (PPC) Trends

As particle count increases, the PPC increases, improving the cache reuse of the grid points during interpolation, leading to better per-particle efficiency.

6.4 Parallel Bottlenecks

Scalability is limited by the serial components of random number generation and the repacking phase in Approach 1. The comparison with Assignment 04 highlights the cost of maintaining a dynamic particle set.

7 Conclusion

The HPC cluster excels at large-scale simulations (10^9 particles) due to high memory capacity, while the Lab PC is faster for small workloads. Dynamic updates introduce significant but manageable performance costs.